Stratified Sampling: (Keeps the Model data distribution but makes the dataset size smaller)

difficulty Analyzer:
- → ① Entropy based Uncertainty
- → ② Spatial complexity → [batch, len, 42, 3]
  - Relu → [0,1]

Valid frames → Hand spread (Hand Variance)

↓

2 hand articulation

↓

Spread within Each hand

↓

Normalize and combine them based on Some x weight (right now are random LLM weights)

↓

then O/p Size between 0 and 1

② Temporal complexity
- → x shape (batch, max len, 42, 3)
- → O/p (0,1)

Valid frames Selection → Velocity (dy/dx) → dy^2/dx^2 (Acceleration of hand)

↓

Mean of Variance avg time of dy/dx and dy^2/dx^2

↓

(direction changes) ⟵

Uses Velocity

data I/p → [t-1, J, D]
- Number of frames from before
- We check the Velocity between Previous and the Next 2 timesteps to see the change in direction.

We get a Matrix then we get cos(θ)
- Angle between Vector
- (−cos(θ)) → direction change

0 → Smooth

Large → Jitter

Take the Average over time for that full Video.

↓

Into some [0,1] based on Some eqn based on LLM weights

④ Motion pattern complexity: (circular, linear, static)

x → (batch, max. len, 42, 3)

Relu → O/p [0,1]

Hand based motion pattern → 2D motion Analysis → Circular vs Linear motion

Mainly Used

① 1DCNN Block: (what happens at that time) → Input (B, dim, Timesteps) ↑ Per dimension convolution

2 types
- → depthwise convolution
- → Pointwise convolutions (channel mixing)

with Batch Normalize + Gelu

Onto a Residual Block (Layer)

② Transformer block for Temporal Reasoning:

↓

B, T, C not B, C, T } Required by transformers

Pre Normalization And why → Read

Self attention → "Each time step which other time step matters"

And then Some

[Linear → Gelu → dropout → Linear]

Then Again Residual block of O/p x (Transposed)

Transpose them back again.

"Transformer" → Which other Moment Explains this Moment